

Alessandro Butti

📍 Milan, Italy ✉ alessandro.butti3@gmail.com

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Professional Summary

Computational neuroscience researcher with training in biomedical engineering and strong Python-based modeling skills. Experience in statistical modeling under missing/censored data, spiking-network simulation, and machine learning optimization.

Research interests include statistical inference for connectomics, mechanistic modeling of cortical circuits, and structure-function relationships in neural systems.

Education

M.Sc. in Biomedical Engineering

Oct 2023 – Mar 2026

Politecnico di Milano, Italy

Specialization in Cell, Tissue and Biotechnologies.

Thesis: *Emergence of Spatially Structured Orientation Tuning in Connectomics-Based Spiking Network Models of V1*

Relevant Coursework (selected): Neuroengineering, Bioengineering of Physiological Control Systems, Bioengineering of Neurosensory Systems, Transport Phenomena in Biological Systems, Methods and Applications of AI in Biomedicine, Computer Science and Elements of Medical Informatics.

B.Sc. in Biomedical Engineering

Oct 2020 – Sep 2023

Politecnico di Milano, Italy

Thesis: *Design and fabrication of a custom syringe pump for microfluidic biological assays.*

Research Experience

Computational Neuroscience Researcher (M.Sc. Thesis)

Sep 2025 – Mar 2026

Politecnico di Milano & Bocconi University

Advisor: Prof. Alberto Antonietti

Co-Advisor: Prof. Alessandro Sanzeni

- Built connectome-informed spiking network models of mouse primary visual cortex (V1) using the MICrONS dataset, integrating synapse-resolution anatomy and functional imaging data.
- Developed probabilistic models for synaptic connectivity reconstruction under missing and censored data; validated using cross-validation on held-out synaptic outputs.
- Analyzed structure-function relationships at microscale level, quantifying orientation tuning statistics and testing the emergence of functional organization from structural connectivity alone.
- Tools: Python, NumPy, Pandas, PyTorch, Optuna, NEST; probabilistic modeling, cross-validation, large-scale simulation pipelines; reproducible analysis workflows.

Manuscript in preparation.

Projects

Deep Learning Classifier for CT Lung Nodule Images

Feb 2025 – Jun 2025

Politecnico di Milano, Course: Applied AI for Biomedicine

Design and implementation of a convolutional neural network (CNN) for multi-class classification of CT lung nodule images across five stages of tumor progression.

Optimization and Biological Validation of 3D Microstructures to Model Cancer Fibrosis Progression

Oct 2024 – Jan 2025

Politecnico di Milano, Course: Cellular Modeling Laboratory

Fabrication of micrometric 3D scaffolds using two-photon polymerization (Nanoscribe) and evaluation of cell colonization to reproduce fibrotic progression in tumor microenvironments.

Augmented Reality Ultrasound (ARUS)

Oct 2024 – Jan 2025

Capstone Project, Politecnico di Milano – in collaboration with EcoCardioService

Development of a functional framework for real-time visualization of echocardiographic videos and images through the Microsoft HoloLens 2 headset. The prototype was successfully tested in a clinical setting.

Technical Skills

Programming: Python (NumPy, Pandas, Matplotlib); PyTorch; TensorFlow; basic C/C++; SQL; MATLAB

Machine Learning & Statistics: supervised learning; probabilistic modeling; cross-validation; hyperparameter optimization (Optuna); deep learning

Modeling & Simulation: spiking neural networks (LIF); large-scale simulation pipelines (NEST)

Tools: Git; HPC; Jupyter

Extracurricular Activities

Several years of experience playing amateur football while pursuing academic studies;

Summer job as a waiter on Lake Como, gaining teamwork and customer service experience;

Experience in tutoring high school students in science subjects;

Participation in Politecnico di Milano events as student staff (Open Days).

Languages

Italian: Native proficiency

English: Upper-intermediate (B2, Cambridge First Certificate)